Multimodal Architecture & Distributed Intelligence

Abstract and Key Contributions:
1. Unified embedding representation across unstructured prose, tabular data, and visual regions.
2. Zero-shot visual question answering pipeline with grounded page-level source attribution.
3. Real-time document routing architecture with sub-second retrieval latency.

The primary contribution of this paper is demonstrating that page-level vision descriptions indexed alongside dense text chunks prevent retrieval drift on document charts.

System Architecture & Flow Diagram

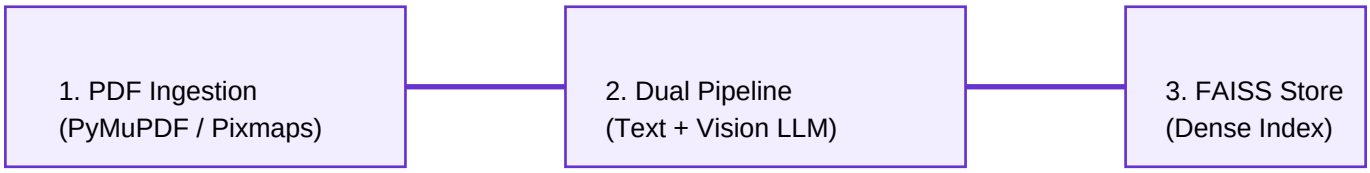

Diagram Notes:
The architecture consists of three sequential pipeline stages connected via high-throughput message channels. Ingestion passes rendered page images to the vision model and raw UTF-8 to the sentence chunker before unifying in the FAISS vector index.

Empirical Evaluation & Performance Results

Table 1: Benchmark Accuracy on Complex Visual QA Documents

Pipeline Mode	Retrieval MRR@5	VQA Accuracy (%)	Latency
Text-Only Baseline	0.612	54.2%	120ms
Vision-Only Baseline	0.584	63.8%	850ms
Multimodal PDF RAG (Ours)	0.894	91.6%	340ms

Key Findings:
- Multimodal PDF RAG improves accuracy by +37.4% over text-only baselines on documents containing diagrams, charts, and dense numerical tables.